AREA DISTORTION AND EXACT BAND GEOMETRY OF THE AM–GM CONE

JEREMY EBERT

Abstract. For the mean-value coordinates

$$X = \frac{x-y}{2}, \qquad Y = \sqrt{xy}, \qquad T = \frac{x+y}{2},$$

which satisfy $X^2 + Y^2 = T^2$, we derive the exact area distortion from the positive factor plane to the flat (X, Y) circle view and then to the induced Euclidean area on the cone. The flat Jacobian is

$$dX\,dY = \frac{x+y}{4\sqrt{xy}}\,dx\,dy = \frac{T}{2Y}\,dx\,dy,$$

while cone surface area is exactly $\sqrt{2}$ times flat-shadow area. Anti-diagonal triangles map to half-disks with global unweighted area ratio $\pi/4$, constant-product hyperbolas $xy = n$ map to horizontal secants $Y = \sqrt{n}$, and the resulting hyperbola/secant partition admits closed formulas on both sides of the map. In logarithmic factor-ratio coordinates, the transformation compactifies the rapidity variable $s = \frac{1}{2}\log(x/y)$ into the ordinary circle angle θ by

$$\sin\theta = \tanh s, \qquad \cos\theta = \operatorname{sech} s, \qquad d\theta = \operatorname{sech} s\,ds.$$

Most importantly, the original factor-plane measure is recovered in circle coordinates by the exact weighted measure

$$dx\,dy = 2\cos\theta\,dX\,dY = 2\operatorname{sech} s\,dX\,dY.$$

Thus the logarithmic, circular, and area descriptions are connected by an exact change of variables rather than by a constant area-preserving identification.

1. The factor-plane map

For $x, y > 0$ define

$$X = \frac{x-y}{2}, \qquad Y = \sqrt{xy}, \qquad T = \frac{x+y}{2}. \tag{1}$$

Then

$$X^2 + Y^2 = T^2, \qquad x = T + X, \qquad y = T - X. \tag{2}$$

The map

$$F : (x, y) \longmapsto (X, Y)$$

is a diffeomorphism from the open positive quadrant onto the open upper half-plane $Y > 0$.

Proposition 1 (Exact flat-area Jacobian). *The factor-plane area element and flat circle-view area element are related by*

$$\boxed{dX\,dY = \frac{x+y}{4\sqrt{xy}}\,dx\,dy = \frac{T}{2Y}\,dx\,dy.} \tag{3}$$

Proof. From (1),

$$X_x = \frac{1}{2}, \quad X_y = -\frac{1}{2}, \quad Y_x = \frac{1}{2}\sqrt{\frac{y}{x}}, \quad Y_y = \frac{1}{2}\sqrt{\frac{x}{y}}.$$

Date: 2026.

Hence
$$\left|\frac{\partial(X,Y)}{\partial(x,y)}\right| = \frac{1}{4}\left(\sqrt{\frac{x}{y}}+\sqrt{\frac{y}{x}}\right) = \frac{x+y}{4\sqrt{xy}} = \frac{T}{2Y}.$$

□

Remark 2. The distortion is not constant. It is smallest on the diagonal $x = y$, where it equals $1/2$, and diverges toward either factor axis. This is the precise obstruction to identifying ordinary multiplication-table area directly with Euclidean area in the circle view.

2. From the circle view to the cone surface

The cone is the graph $T = \sqrt{X^2+Y^2}$ over the upper half-plane. Away from the apex,
$$|\nabla T|^2 = 1,$$
so its Euclidean surface element is

$$\boxed{dA_{\text{cone}} = \sqrt{2}\,dX\,dY.} \tag{4}$$

Combining (3) and (4) gives

$$\boxed{dA_{\text{cone}} = \frac{x+y}{2\sqrt{2}\sqrt{xy}}\,dx\,dy = \frac{T}{\sqrt{2}\,Y}\,dx\,dy.} \tag{5}$$

This is the induced-area formula derived independently in the earlier semiclassical note; here it is used purely geometrically.

Proposition 3 (Exact anti-diagonal band area)**.** *The factor strip*
$$K \le x+y \le K+2$$
maps to the half-annulus with radii $T_1 = K/2$ and $T_2 = (K+2)/2$. Its flat-shadow area is

$$A_{\text{shadow}}(K) = \frac{\pi}{2}(T_2^2 - T_1^2) = \frac{\pi}{2}(K+1), \tag{6}$$

and its induced cone-surface area is

$$\boxed{A_{\text{band}}(K) = \frac{\pi}{\sqrt{2}}(K+1).} \tag{7}$$

3. Whole triangles map to half-disks

For $S > 0$ let
$$\Delta_S = \{(x,y) : x \ge 0,\ y \ge 0,\ x+y \le S\}.$$
Since $T = (x+y)/2$, its image is the upper half-disk
$$H_S = \{(X,Y) : Y \ge 0,\ X^2+Y^2 \le (S/2)^2\}.$$
Therefore
$$\text{Area}(\Delta_S) = \frac{S^2}{2}, \qquad \text{Area}(H_S) = \frac{\pi S^2}{8}.$$

Corollary 4 (Global triangle-to-half-disk factor)**.** *Although the local Jacobian (3) is nonuniform, every complete anti-diagonal triangle satisfies the exact unweighted Euclidean relation*

$$\boxed{\text{Area}(H_S) = \frac{\pi}{4}\,\text{Area}(\Delta_S).} \tag{8}$$

4. The constant-product hyperbola becomes a secant

Fix $n > 1$ and use the smallest continuous anti-diagonal shell containing the extremal factor points $(1, n)$ and $(n, 1)$:

$$S = n + 1.$$

Inside Δ_{n+1} the constant-product boundary

$$xy = n$$

meets the shell at $(1, n)$ and $(n, 1)$. Under F it becomes

$$\boxed{Y = \sqrt{n},} \tag{9}$$

a horizontal secant of the half-disk of radius

$$R = \frac{n+1}{2}.$$

Its half-width is

$$a = \frac{n-1}{2},$$

and the right-triangle identity is

$$\boxed{\left(\frac{n-1}{2}\right)^2 + n = \left(\frac{n+1}{2}\right)^2.} \tag{10}$$

Let B_n be the portion of Δ_{n+1} satisfying $xy \le n$, and let C_n be its complement in the triangle. Since (9) is exact,

$$F(B_n) = \{(X, Y) \in H_{n+1} : 0 \le Y \le \sqrt{n}\},$$

and

$$F(C_n) = \{(X, Y) \in H_{n+1} : Y \ge \sqrt{n}\}.$$

Thus the hyperbola partition and the secant partition are the same partition transported through the nonuniform Jacobian (3).

Proposition 5 (Factor-plane areas of the two regions)**.** *The factor-plane areas are*

$$\text{Area}(B_n) = n \log n + n + 1, \tag{11}$$

$$\text{Area}(C_n) = \frac{n^2 - 1}{2} - n \log n. \tag{12}$$

Proof. For $1 \le x \le n$, the lower boundary is $0 \le y \le n/x$, contributing

$$\int_1^n \frac{n}{x}\, dx = n \log n.$$

For $0 \le x \le 1$ and $n \le x \le n + 1$, the shell line $y = n + 1 - x$ lies below the hyperbola, so the two outer wings contribute

$$\int_0^1 (n + 1 - x)\, dx + \int_n^{n+1} (n + 1 - x)\, dx = n + 1.$$

This gives (11), and subtracting from the full triangle area $(n + 1)^2/2$ gives (12). $\square$

Remark 6 (Where $n \log n$ sits). The classical area $n \log n$ is the central under-hyperbola region with $1 \le x \le n$. The complete shell region below the hyperbola has an additional pair of wings of total area $n + 1$. This distinction is necessary when comparing the continuous hyperbola picture with the entire enclosing half-disk.

5. Exact circular cap and below-secant areas

Define the half central angle θ_n by

$$\cos\theta_n = \frac{\sqrt{n}}{R} = \frac{2\sqrt{n}}{n+1}. \tag{13}$$

Equivalently,

$$\sin\theta_n = \frac{n-1}{n+1}, \qquad \tan\theta_n = \frac{n-1}{2\sqrt{n}}. \tag{14}$$

Proposition 7 (Secant partition of the half-disk)**.** *The flat circular cap above $Y = \sqrt{n}$ has area*

$$\boxed{A^\circ_{\mathrm{cap}}(n) = \frac{(n+1)^2}{4}\,\theta_n - \frac{(n-1)\sqrt{n}}{2}.} \tag{15}$$

The complementary region below the secant has area

$$\boxed{A^\circ_{\mathrm{below}}(n) = \frac{\pi(n+1)^2}{8} - A^\circ_{\mathrm{cap}}(n).} \tag{16}$$

Proof. The cap is a circular sector of central angle $2\theta_n$ minus the isosceles triangle with half-base $(n-1)/2$ and height $\sqrt{n}$. Thus its area is

$$R^2\theta_n - \frac{(n-1)\sqrt{n}}{2},$$

which is (15). Equation (16) follows from the half-disk area $\pi R^2/2$. □

Corollary 8 (Exact push-forward of the hyperbola partition)**.** *Under the Jacobian (3),*

$$\boxed{n\log n + n + 1 \quad \xrightarrow{F} \quad A^\circ_{\mathrm{below}}(n),} \tag{17}$$

and

$$\boxed{\frac{n^2-1}{2} - n\log n \quad \xrightarrow{F} \quad A^\circ_{\mathrm{cap}}(n).} \tag{18}$$

The arrows denote image regions under a non-area-preserving map, not equality of the displayed numbers.

Proposition 9 (Push-forward of the literal $n\log n$ region)**.** *Let*

$$E_n = \left\{(x,y) : 1 \le x \le n,\ 0 \le y \le \frac{n}{x}\right\}.$$

Then $\mathrm{Area}(E_n) = n\log n$*, while the flat area of its image is*

$$\boxed{\mathrm{Area}(F(E_n)) = \frac{2}{3}(n-1)\sqrt{n}.} \tag{19}$$

Proof. Use (3):

$$\mathrm{Area}(F(E_n)) = \int_1^n \int_0^{n/x} \frac{x+y}{4\sqrt{xy}}\,dy\,dx.$$

The inner integral equals

$$\frac{\sqrt{n}}{2} + \frac{n^{3/2}}{6x^2}.$$

Integrating from 1 to n gives

$$\frac{n-1}{2}\sqrt{n} + \frac{n-1}{6}\sqrt{n} = \frac{2}{3}(n-1)\sqrt{n}.$$

□

6. Logarithmic rapidity and circular angle

The same map has a particularly simple expression in product-ratio coordinates. Write

$$\rho = \sqrt{xy}, \qquad s = \frac{1}{2}\log\frac{x}{y}. \tag{20}$$

Then

$$x = \rho e^{s}, \qquad y = \rho e^{-s}, \tag{21}$$

and therefore

$$\boxed{X = \rho\sinh s, \qquad Y = \rho, \qquad T = \rho\cosh s.} \tag{22}$$

Thus the constant-product curve $xy = \rho^2$ is a horizontal line in the (X, Y) view, while s is exactly its hyperbolic rapidity coordinate.

Let θ denote ordinary polar angle measured from the positive Y axis, so

$$X = T\sin\theta, \qquad Y = T\cos\theta.$$

Comparing with (22) gives the exact dictionary

$$\boxed{\sin\theta = \tanh s, \qquad \cos\theta = \operatorname{sech} s, \qquad \tan\theta = \sinh s.} \tag{23}$$

Differentiating yields

$$\boxed{d\theta = \operatorname{sech} s\, ds.} \tag{24}$$

Consequently

$$\boxed{\theta(s) = \int_0^s \operatorname{sech} t\, dt,} \tag{25}$$

and the full positive rapidity ray is compactified to a quarter turn:

$$\boxed{\int_0^\infty \operatorname{sech} t\, dt = \frac{\pi}{2}.} \tag{26}$$

Since $\operatorname{sech} t = 2/(e^t + e^{-t})$, equation (26) is the exact place in this geometry where an unbounded logarithmic/exponential coordinate is converted into a bounded circular angle involving π.

For the shell endpoint $(x, y) = (n, 1)$,

$$s_n = \frac{1}{2}\log n, \tag{27}$$

and (23) becomes precisely (13)–(14):

$$\boxed{\theta_n = \arctan\left(\sinh\frac{1}{2}\log n\right) = \arccos\left(\operatorname{sech}\frac{1}{2}\log n\right).} \tag{28}$$

Remark 10 (Area elements in product-ratio coordinates)*.* From (21),

$$dx\, dy = 2\rho\, d\rho\, ds,$$

whereas (22) gives

$$dX\, dY = \rho\cosh s\, d\rho\, ds.$$

Their ratio is $\cosh s/2$, exactly the Jacobian $T/(2Y)$ in (3). Thus the same hyperbolic factor $\cosh s$ controls the local area distortion, while its reciprocal $\operatorname{sech} s$ controls the compression from rapidity to circle angle.

7. The exact transported factor-area measure

The failure of a constant area normalization has a simple exact resolution.

Proposition 11 (Weighted circle measure)**.** *Let θ be measured from the positive Y axis. Then*

$$\boxed{dx\,dy = 2\cos\theta\,dX\,dY.} \tag{29}$$

Equivalently,

$$\boxed{dx\,dy = 2T\cos\theta\,dT\,d\theta.} \tag{30}$$

In rapidity variables,

$$\boxed{dx\,dy = 2\,\mathrm{sech}\,s\,dX\,dY.} \tag{31}$$

On the cone surface,

$$\boxed{dx\,dy = \sqrt{2}\cos\theta\,dA_{\mathrm{cone}}.} \tag{32}$$

Proof. Equation (3) gives

$$dx\,dy = \frac{2Y}{T}\,dX\,dY.$$

Since $Y/T = \cos\theta = \mathrm{sech}\,s$, (29) and (31) follow. In polar circle coordinates $dX\,dY = T\,dT\,d\theta$, giving (30). Finally use $dA_{\mathrm{cone}} = \sqrt{2}\,dX\,dY$ to obtain (32). □

Corollary 12 (Exact recovery of arbitrary factor-plane regions)**.** *For every measurable factor-plane region E,*

$$\boxed{\mathrm{Area}_{xy}(E) = \iint_{F(E)} 2\cos\theta\,dX\,dY.} \tag{33}$$

Thus the original factor-plane measure is represented exactly in the circle view by the angular weight $2\cos\theta$.

For the literal logarithmic region E_n of Proposition 9, this yields

$$\boxed{n\log n = \iint_{F(E_n)} 2\cos\theta\,dX\,dY.} \tag{34}$$

The raw circle-image area $(2/3)(n-1)\sqrt{n}$ and the weighted area $n\log n$ are therefore two distinct exact measures of the same transported region.

The whole-triangle relation provides a check. If $R = S/2$, then

$$\int_{-\pi/2}^{\pi/2}\int_0^R 2T\cos\theta\,dT\,d\theta = 2R^2 = \frac{S^2}{2},$$

recovering the factor-triangle area exactly. The unweighted half-disk area $\pi R^2/2$ gives the separate global ratio $\pi/4$ of Corollary 4.

8. The $n = 11$ shell

For $n = 11$,

$$R = 6, \qquad Y = \sqrt{11}, \qquad a = 5,$$

and

$$\theta_{11} = \arccos\frac{\sqrt{11}}{6} = \arctan\frac{5}{\sqrt{11}}.$$

The upper half-disk has raw Euclidean area

$$18\pi,$$

while

$$A^{\circ}_{\mathrm{cap}}(11) = 36 \arccos \frac{\sqrt{11}}{6} - 5\sqrt{11},$$
$$A^{\circ}_{\mathrm{below}}(11) = 18\pi - A^{\circ}_{\mathrm{cap}}(11).$$

Numerically,

$$A^{\circ}_{\mathrm{cap}}(11) \approx 18.880864, \qquad A^{\circ}_{\mathrm{below}}(11) \approx 37.667804.$$

The literal $11 \log 11$ region has raw image area

$$W_L(11) = \frac{20}{3}\sqrt{11} \approx 22.110832,$$

but its weighted circle measure is exactly

$$11 \log 11 \approx 26.376848.$$

The discrete shell values

$$T_{11} = 66, \qquad D(11) = 29, \qquad A_{11} = 37, \qquad 11H_{11} \approx 33.218651$$

remain discrete staircase quantities and are not identified with the raw circular areas.

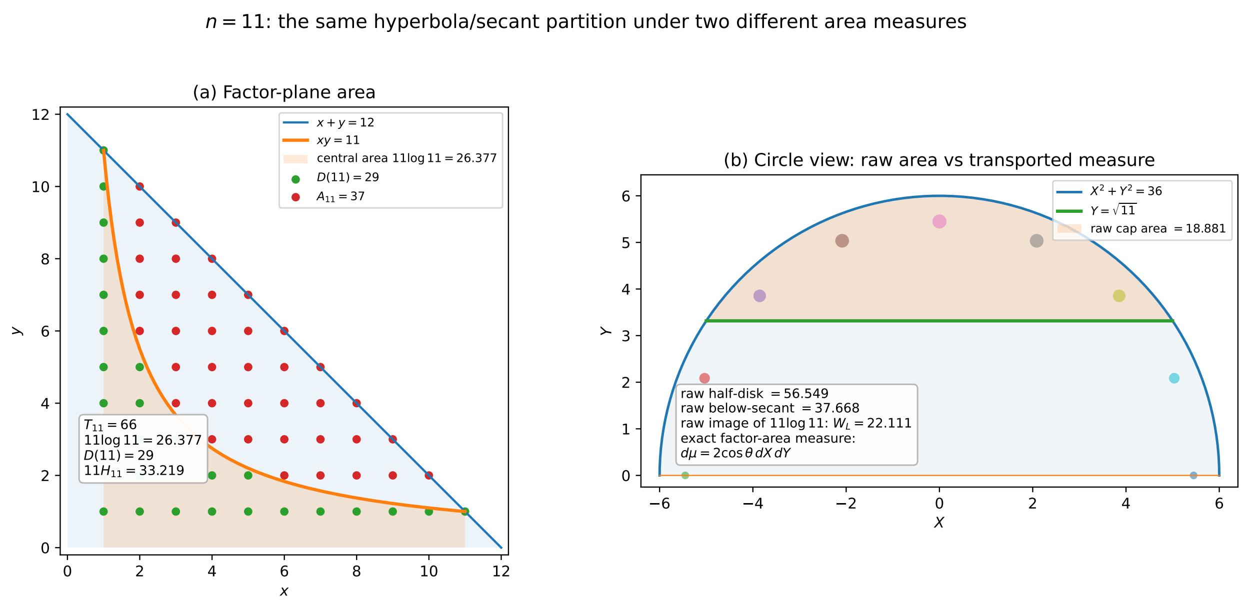

FIGURE 1. The $n = 11$ hyperbola/secant partition under two area measures. Left: the factor-plane triangle and the classical $xy = 11$ boundary, with the continuous $11 \log 11$ region and the discrete $D(11)/A_{11}$ partition shown together. Right: the same continuous boundary is the secant $Y = \sqrt{11}$ of the radius-6 half-disk. Raw Euclidean circle area and transported factor-plane area are distinct: the exact factor-area measure in the circle view is $d\mu = 2\cos\theta \, dX \, dY$.

9. Conclusion

The factor-plane triangle, the flat circle view, and the Euclidean cone surface carry three distinct but exactly related area measures. The first change of variables is nonuniform and is governed by $T/(2Y)$; the second is the constant factor $\sqrt{2}$. Constant-product hyperbolas become secants, and the complete shell partition has closed formulas in both coordinate systems. In product-ratio variables the same transformation sends logarithmic rapidity s to ordinary circle angle θ through

$d\theta = \operatorname{sech} s\, ds$, providing an exact logarithmic-to-circular compactification. The factor-plane area measure itself is recovered in circle coordinates by the canonical weight $2\cos\theta = 2\operatorname{sech} s$. This weighted measure, rather than any global constant normalization, is the exact bridge needed for comparing factor-plane areas with their circular images.

References

[1] J. Ebert, *The Cutting Plane: Every Line Through the Multiplication Table Is a Conic Section*, 2026.
[2] J. Ebert, *A Semiclassical Band-Area Correspondence for the AM–GM Cone*, research note, 2026.

Franklin County, Ohio, USA